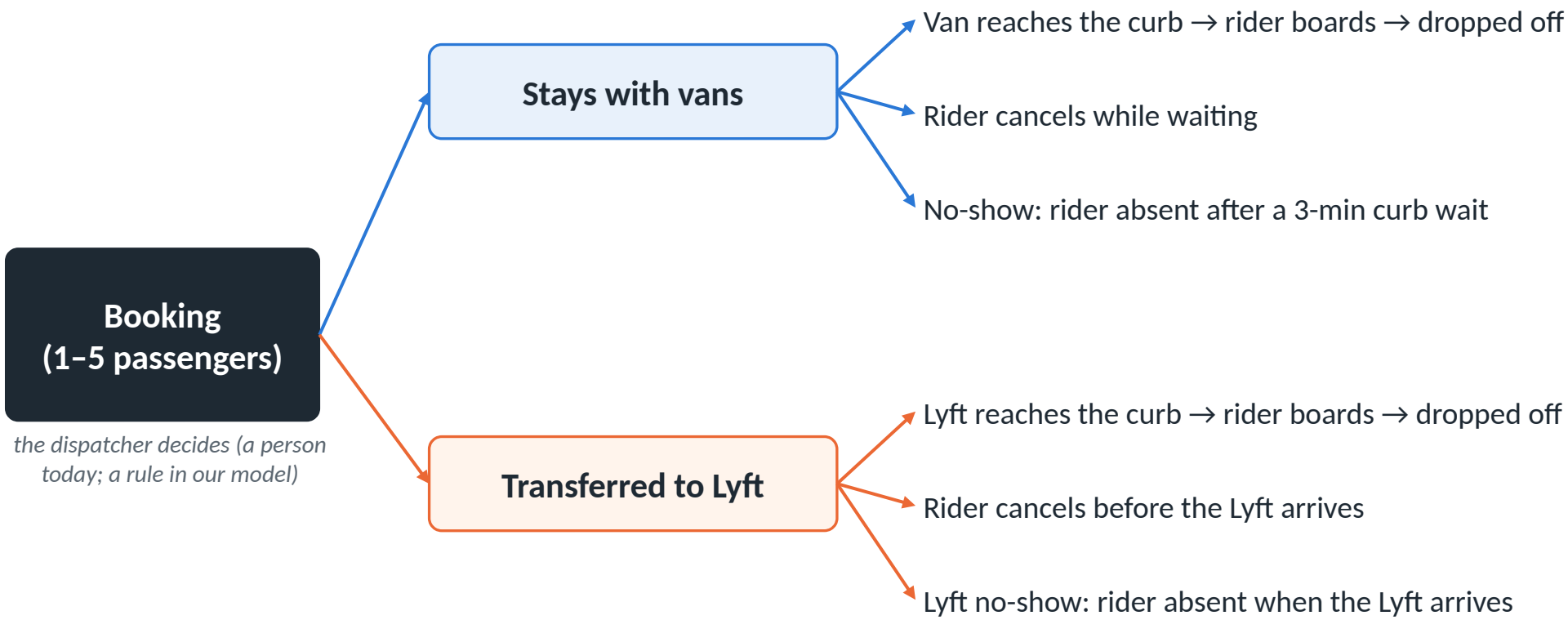
# Testing Lyft transfer decisions for Blue Jay Night Ride

What aggregate data can tell us · how we built synthetic evenings · what replaying them showed

Team FourSight · EN.553.602 Research and Design in Mathematics · Midterm, October 2026
Content draft v0.3 · 20 min presentation + 10 min questions · speakers to be assigned

# One booking, several possible outcomes




**Booking**
one request in the app

**Passenger**
one person in a booking

**Curb arrival**
the vehicle reaches the pickup point

**Pickup**
the rider boards

**Cancellation**
the rider gives up before pickup

## Our question: when should a waiting booking be moved to Lyft?

Source: JHU Night Ride service pages (3-min curb wait); scenario_spec.md §5–§6.

# What we originally proposed

## September proposal: relied on trip-level records

1. Replay real service nights
2. Compare simple rules with an optimization model re-solved as requests arrive
3. Choose Lyft overflow inside that optimizer
4. Treat published service targets as hard limits

## Three proposal numbers we corrected

~~“About 5,100 rides per hour at 6 pm”~~

→ a 30-day total: about 170 in one evening's 6 pm hour

~~“About 28 vans”~~

→ the “All services” average (27.6), not a Homewood-only count

~~“1 in 4 trips overflow to Lyft”~~

→ 24.5% of riders rode Lyft: a share of riders, not of trips

Corrected claims and dates: backup B9 →

Source: BlueJayNightRide_Proposal.pptx (September); April 2026 dashboards; methodology v1.4 §1a.

# What April's dashboards actually contain

## Totals and shares (April 2026)

- ✓ **Bookings by status**
  50,297 in the month
- ✓ **Completed rides by hour**
  about 5,100 at 6 pm, summed over 30 days
- ✓ **Vehicles on duty by hour**
  27.6 at 6 pm, all services
- ✓ **Pickup and drop-off maps**
  counts per place, 188 places
- ✓ **Wait-time percentiles**
  per month and hour
- ✓ **Riders by van and by Lyft**
  44,374 riders, 24.5% by Lyft

## Never a single journey

- ✗ **Who asked, and when**
- ✗ **From where to where**
- ✗ **How long that rider waited**
- ✗ **What happened next: boarded, gave up, moved to Lyft**

*So before testing any policy, we had to work out what the totals meant.*

Source: April 2026 operations dashboards (restricted; digitized aggregates in data/processed); methodology v1.4 §3.

# First, we investigated the numbers

## How many requests does a van face, and how many riders each?

**≈ 5,100**

completed rides at 6 pm (30-day total)

→ slides 6 and 8

## Which pickups go with which drop-offs?

**188**

map places with pickup and drop-off counts only

→ slide 7

## What do “Canceled” and “No Show” count?

**23,843**

bookings ended “Canceled”

→ slides 9–10

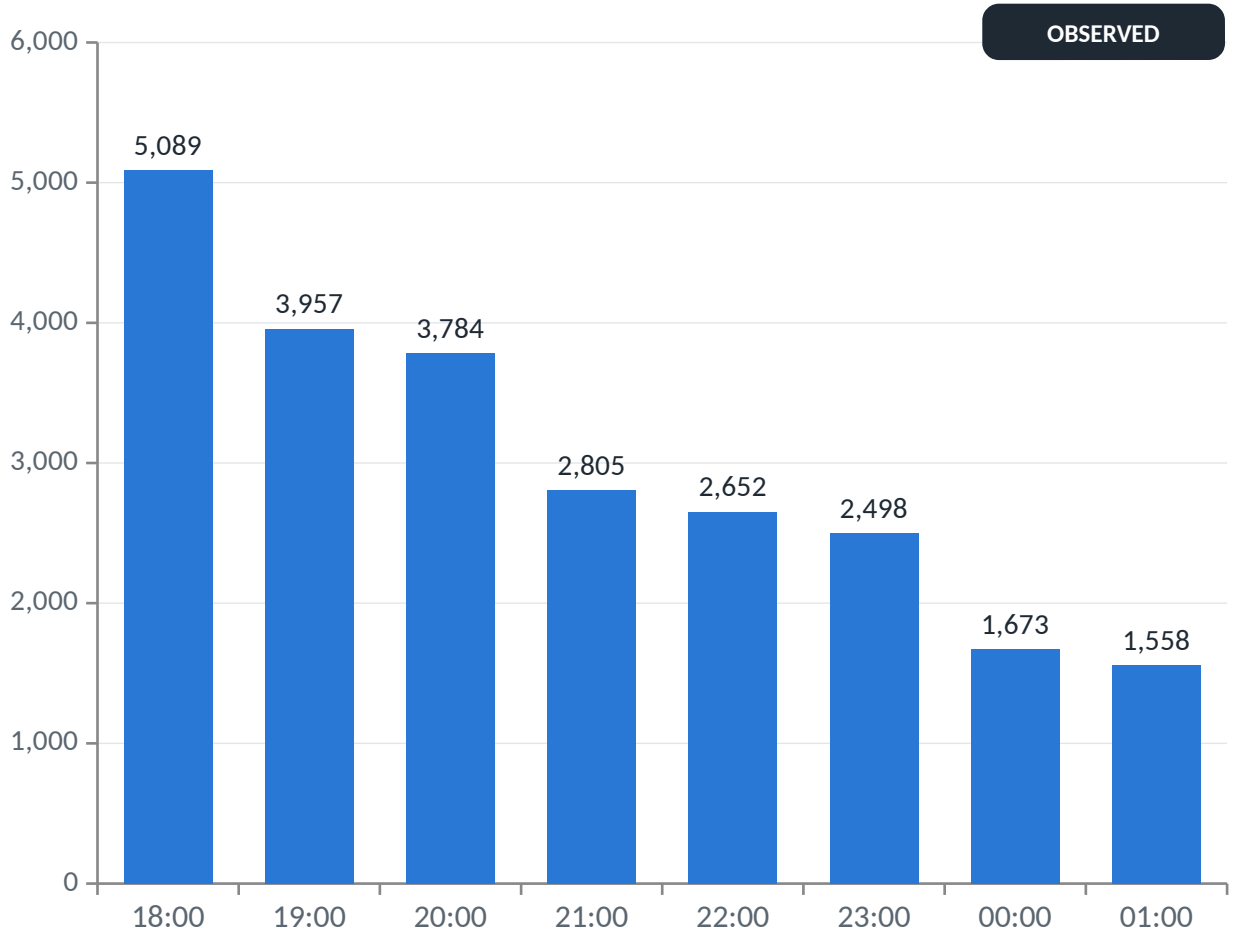
*Each answer supplied one part of a synthetic evening, and exposed one assumption.*

Labels used from here on: OBSERVED read from April dashboards · DERIVED computed from observed numbers · ASSUMED our choice · DESIGNED hand-built test evening · SIMULATED output of our model

Source: April 2026 dashboards (digitized aggregates).

# How many requests should our scenario contain?

Completed rides by hour, summed over 30 April days (digitized)




≈ 5,100 ÷ 30 **≈ 170** DERIVED

completed rides in one evening's 6 pm hour

× 2.09 **≈ 355** DERIVED

bookings in that hour (2.09 bookings per completed ride)

÷ 27.6 **≈ 12.85** DERIVED

bookings per vehicle per hour at 6 pm (unrounded inputs)

× 5 vans **64 / 50 / 54** DERIVED

bookings per hour, first three hours of a five-van evening: 168 in all (each hour uses its own rides and vehicles)

Unknown: how many vehicles served Homewood, and how many bookings were repeats. The pilot (slide 16) uses 2 vans and 28 bookings in one hour.

Source: April dashboards: rides by hour, status totals, vehicles on duty; decision log D-35, D-36, D-49.

# Pickup and drop-off maps do not identify trips

## Same place totals, different trips

### Trips 1

| | to X | to Y | total |
|---|---|---|---|
| **from A** | 2 | 0 | **2** |
| **from B** | 0 | 2 | **2** |
| **total** | **2** | **2** | **4** |

### Trips 2

| | to X | to Y | total |
|---|---|---|---|
| **from A** | 1 | 1 | **2** |
| **from B** | 1 | 1 | **2** |
| **total** | **2** | **2** | **4** |

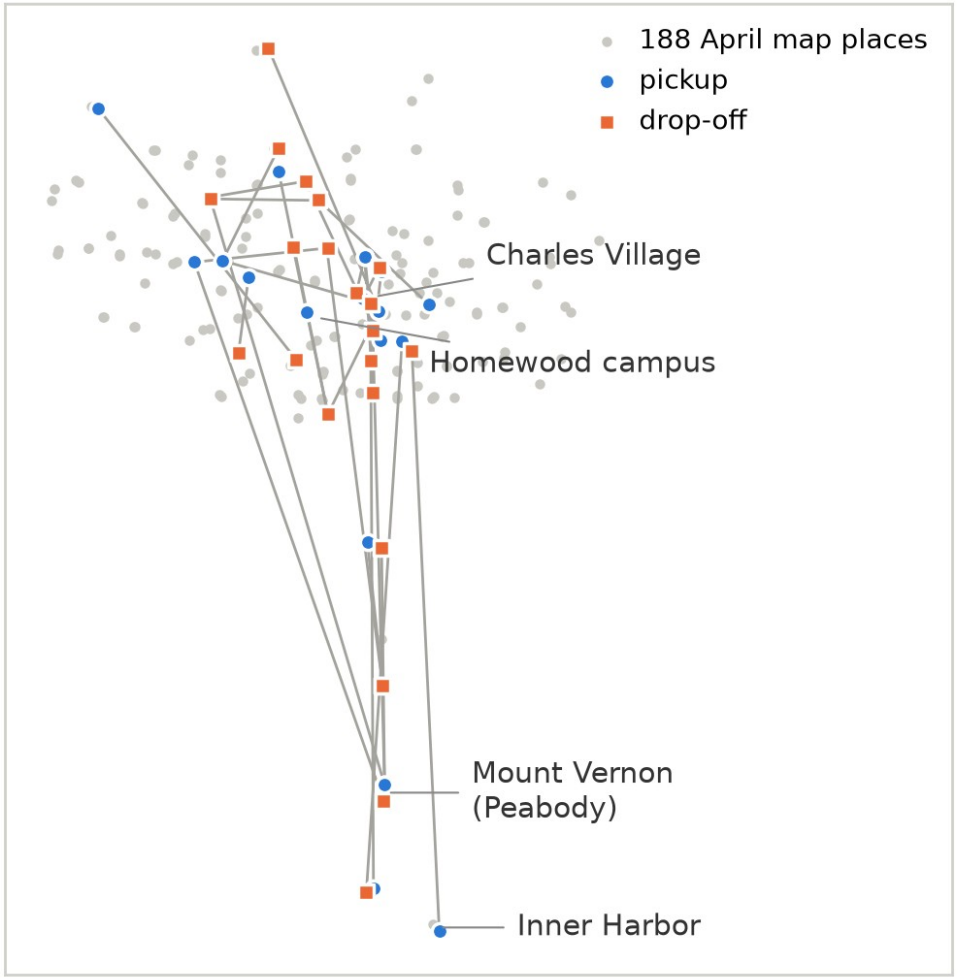
Green cells are all the maps show. Both tables match them exactly.

ASSUMED

### Our pairing choice

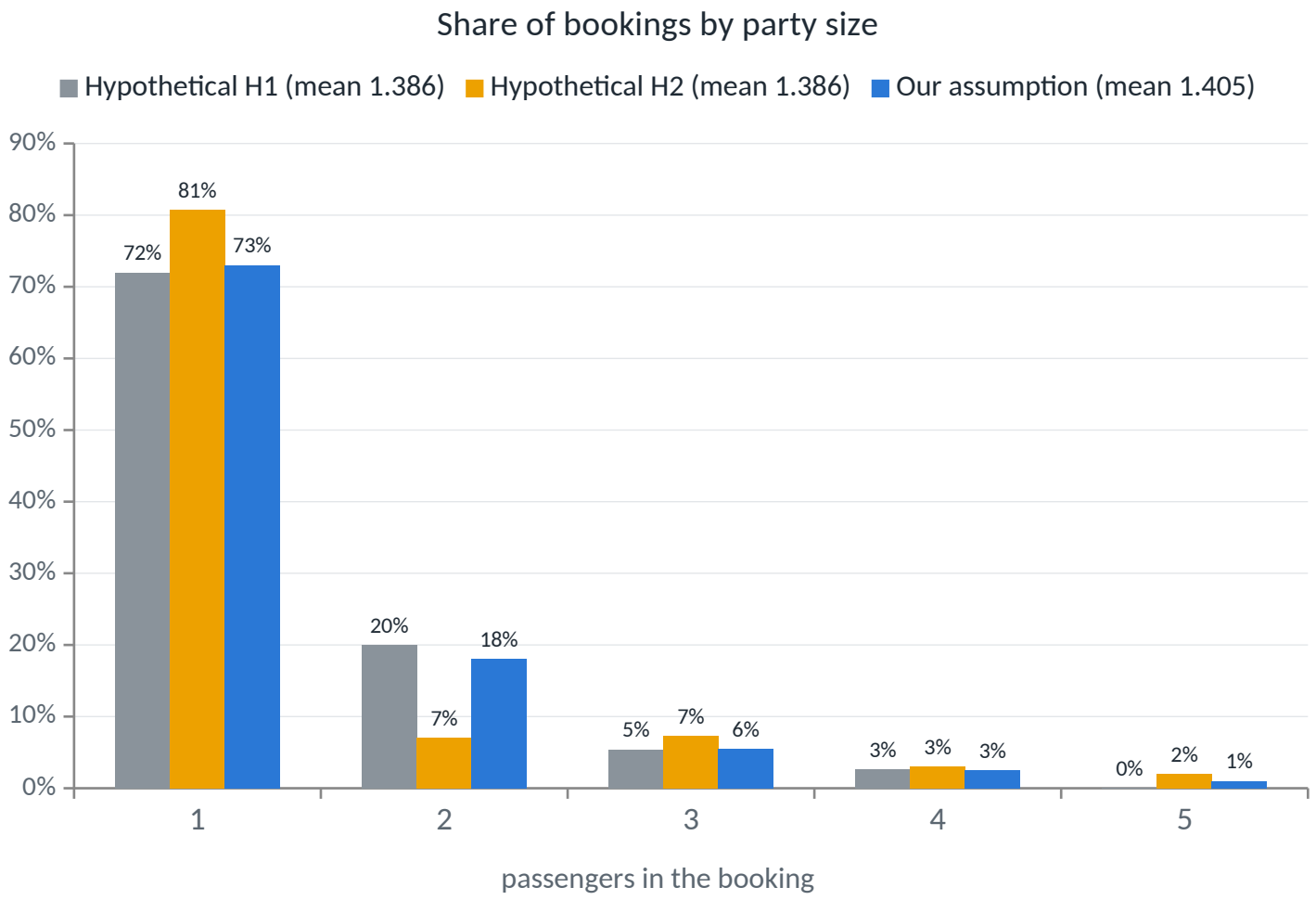
- Nearby trips more likely: weight shrinks about 10% per extra minute of driving
- Then scaled until both map totals match exactly
- The shrink rate is our choice; the data cannot identify it

One synthetic hour: 28 bookings




Source: April pickup and drop-off maps (place totals); decision log D-18; map: the first synthetic pilot night.

# An average group size leaves many possibilities




DERIVED

**1.386**

van passengers per completed van booking (ratio of two observed totals; counts only completed van bookings)

## Bookings with 3 or more riders

| 8% | 12.3% | 9% |
|---|---|---|
| H1 | H2 | Ours |

ASSUMED

- Same mean, different seat needs
- Ours covers all bookings, so its mean (1.405) need not equal April's
- 1% parties of five; they need a larger Lyft
- Bar labels rounded

Source: April dashboards (riders, completed van bookings); decision log D-38, D-33. H1 and H2 are illustrations, not data.

# What does “Canceled” mean?




## 23,843 of 50,297 bookings (47%) ended “Canceled”

### One label, two different events

**Rider gave up**
assumed 50–75% of “Canceled” (a range to test)

**Moved to Lyft by the dispatcher**
assumed to close as “Canceled” in the van system (not confirmed)

*The data cannot separate the two.*

ASSUMED

### Patience is a hidden deadline

- To let riders give up in the model, each rider gets a personal deadline the dispatcher cannot see
- Placeholder: median 18 minutes, until calibrated
- Results depend on it (slide 19)

Source: April dashboard status totals (the digitized sums also show 99 “Denied”); decision log D-39, D-49.

# Why the no-show denominator matters

### Divide by all bookings

$$\frac{2{,}359}{50{,}297} = 4.7\%$$

Under our status mapping, a cancelled booking never had a vehicle at the curb

### Divide by van arrivals (completed + no-show)

$$\frac{2{,}359}{26{,}454} = 8.9\%$$

A no-show can only happen when a vehicle arrives

DERIVED

ASSUMED

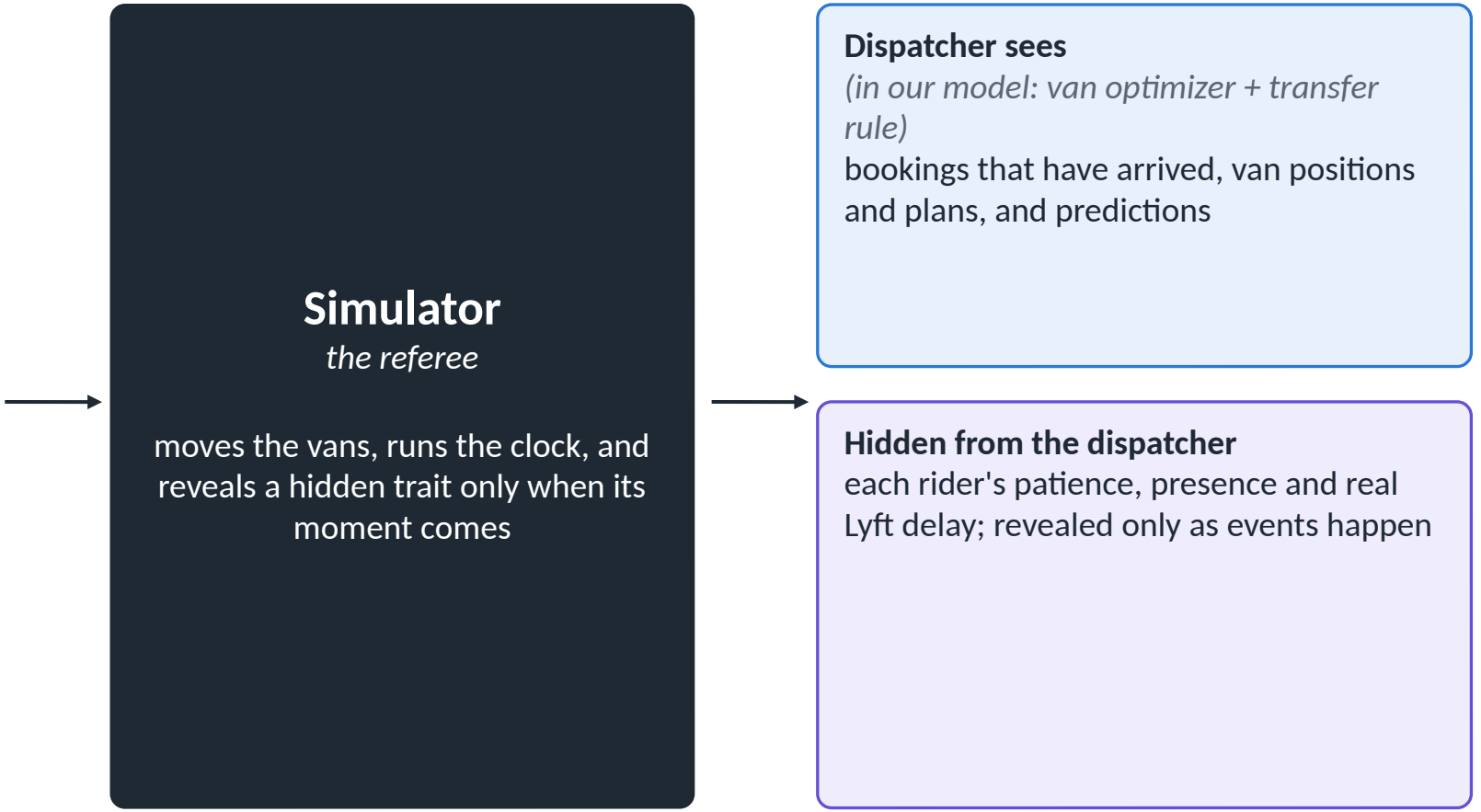
We use 8.9% per vehicle arrival, provisionally. It holds only if “No Show” was recorded after a real van arrival and “Completed” counts van trips. The same count gives two rates, depending on what you divide by.

Source: April dashboard status totals; decision log D-40, D-48; methodology v1.4 Appendix C.

# Our synthetic evening is assembled

## Inputs

| | |
|---|---|
| Bookings: time, place, party size | DERIVED |
| Trip pairing | ASSUMED |
| Vans on duty | DERIVED |
| Travel times: road network × 1.2 | DERIVED |
| Hidden: patience deadline | ASSUMED |
| Hidden: present at the curb? | DERIVED |
| Hidden: Lyft delay | ASSUMED |




Two kinds of evening: small designed ones with hand-computed answers, and randomized one-hour nights for the pilot.

Inputs: B4 → Event order: B3 →

Source: scenario_spec.md v0.8 §1, §17; methodology v1.4 §2–§3.

# How the van optimizer chooses a plan

1 Reveal and update → 2 Generate feasible routes → 3 Select compatible routes → 4 Evaluate transfer rule → 5 Apply accepted transfers → 6 Re-plan once if needed → 7 Execute → 8 Repeat

every 2 minutes and at each new event · blue = route optimizer · orange = transfer rule · grey = simulator

## Step 3: a mixed-integer linear program (MILP) picks one route per van

ILLUSTRATIVE One van; rider 1 has waited 8 min; a party of two has waited 1 min

| Candidate | Waiting | Riding | Driving | Points |
|---|---|---|---|---|
| Route A: rider 1 first | 52.5 | 13.0 | 1.0 | **66.5 ✓ chosen** |
| Route B: party of two first | 67.5 | 12.5 | 1.0 | **81.0** |
| Postpone a booking | priced at its earliest physically possible pickup, plus a large constant so the plan serves as many bookings as the candidates allow | | | |

In Route B rider 1 waits 16 min; past 15 min each minute costs 10 points.

### Points, not money

- Waiting, per passenger-minute: 1 point for the first 5 min, then 2.5, 5, 10 and 20 after 5, 10, 15 and 20 min
- Riding: 0.5 per passenger-minute; driving: 0.1 per km
- Hard limits: seats; pickup before drop-off
- Ride-time guardrail max(25 min, 2 × direct + 5): no plan may create or worsen an excess

"Optimal" = best among the routes we generated at that moment (up to 3 new pickups per van), proven by the solver. Not over every possible route, and not over the whole evening.

Formulation: B2 →

Source: model_design.md v0.5 §3–§4; methodology v1.4 §6 (worked example, boarding time omitted); decision log D-29, D-47, D-52.

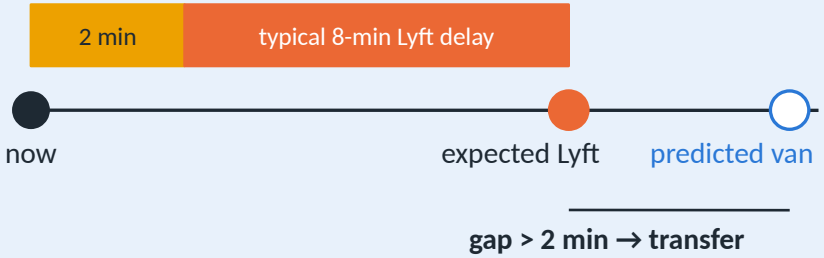
# The six transfer decisions we tested

| Rule | What it reads | Moves the booking to Lyft when |
|---|---|---|
| **Vans only** | — | never |
| **Wait ten minutes** | time already waited | ≥ 10 min |
| **Wait fifteen minutes** | time already waited | ≥ 15 min |
| **Unassigned for ten minutes** | time with no van planning to serve it | ≥ 10 min |
| **Predicted total wait exceeds fifteen minutes** | time waited + van's predicted pickup | > 15 min |
| **Lyft arrives sooner** | predicted van pickup vs expected Lyft | Lyft more than 2 min sooner |

grey: waiting-time rules · green: assignment rule · blue: prediction rules. Only bookings still waiting, with no van already driving to them.

Code key: B1 →

## "Lyft arrives sooner", at one decision moment



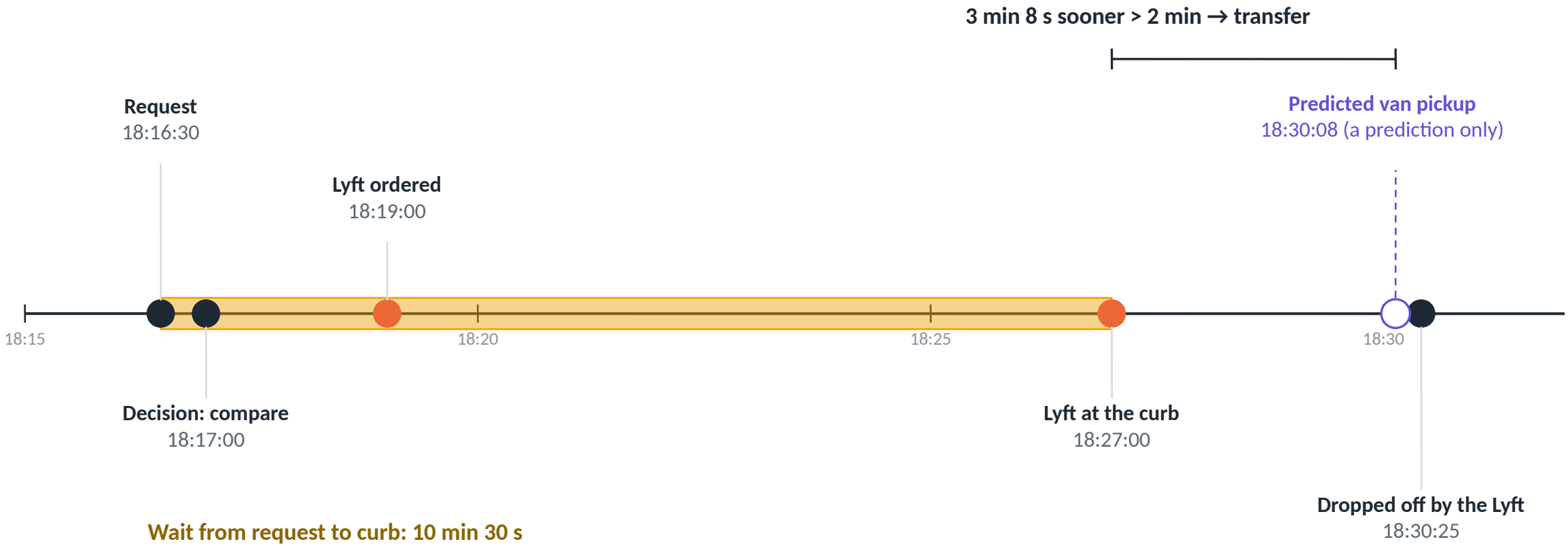

- Expected Lyft pickup = now + 2-min ordering lag (time to place the Lyft order, assumed) + typical 8-min delay
- Transfer only if that is more than 2 minutes sooner than the van plan's predicted pickup
- Both are predictions; the real Lyft delay is hidden

Source: scenario_spec.md v0.8 §6; src/sim/lyft.py. Thresholds fixed before any run.

# One request: compare van versus Lyft

DESIGNED

A small designed evening, separate from the randomized pilot. Rule: “Lyft arrives sooner”.



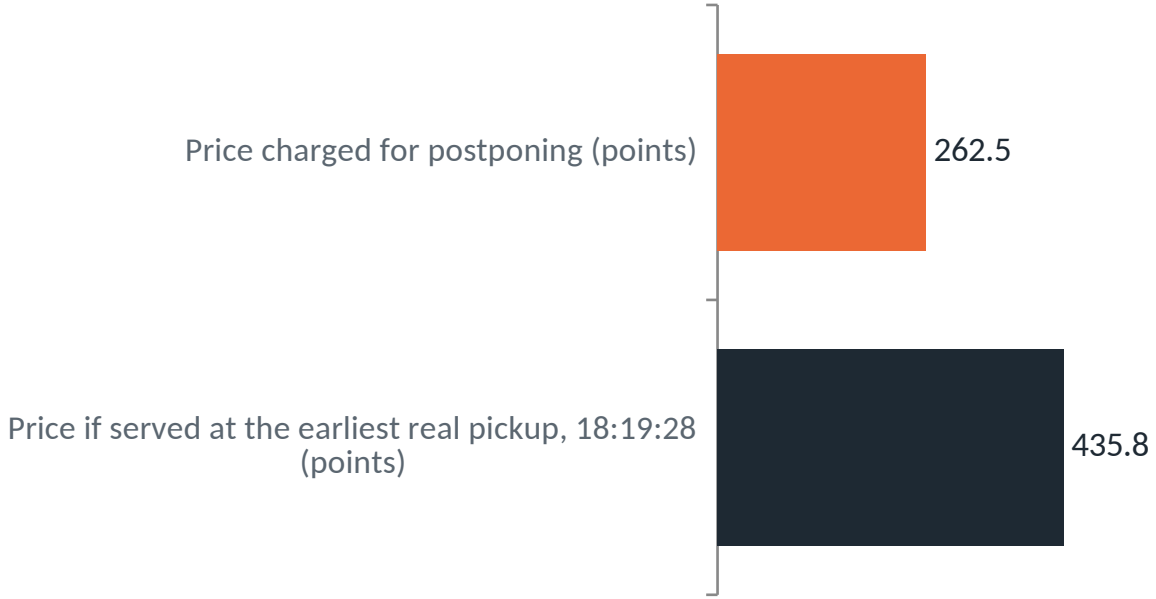

At 18:17:00: expected Lyft = 18:17:00 + 2-min ordering lag + 8-min typical delay = 18:27:00. On this designed evening the delay is fixed, so the Lyft arrived exactly when expected. What the van would have done is never observed.

Source: outputs/sim/L4_R3-2 (epochs.jsonl, outcomes.csv: request R011); scenario_spec.md §6.

# Testing exposed an optimistic postponement price

*A check on the optimizer before we compare rules*

**1 Underpricing: at 18:15 on the designed evening, a party of five has waited 15 min**




Postponing looked cheaper than any service a van could really give, so the longest waiters were left waiting.

**2 Controlled comparison: whole designed evening, vans only (third version: B5)**

| | First version | Chosen |
|---|---|---|
| Longest wait | 44.4 min | **35.6 min** |
| Mean wait | 18.8 min | 18.8 min |
| Median wait | 17.1 min | 18.6 min |
| Extra variables | 0 | 0 |

Waiting was redistributed, not reduced on average.

CHOSEN

3 Price the earliest physically possible pickup

4 Still optimistic: ignores the nearest van's other work

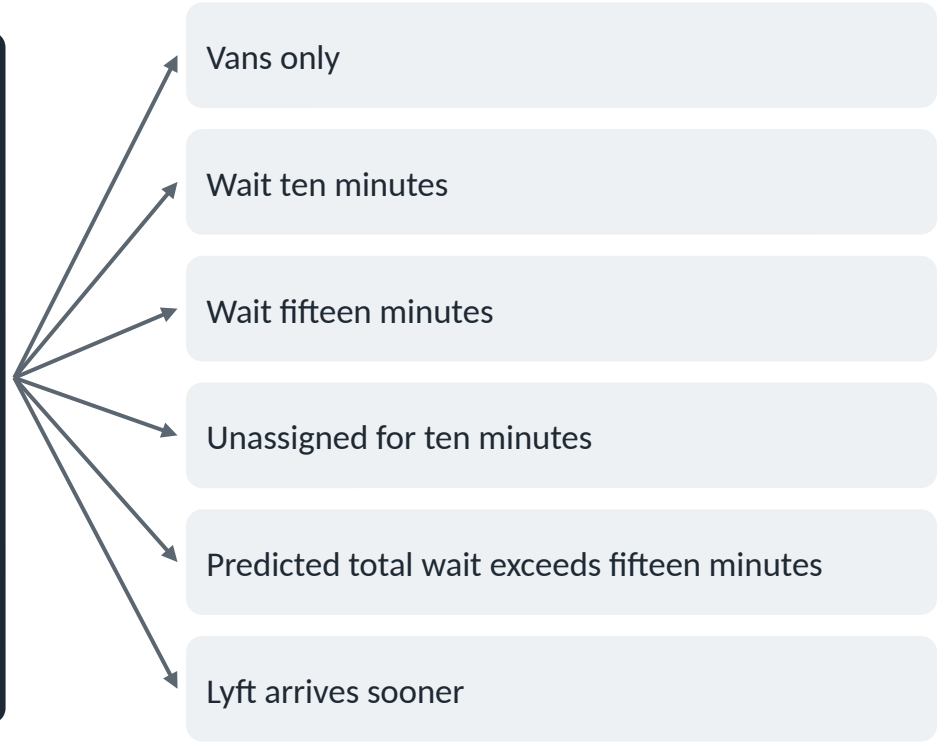
Three formulations compared: B5 →

Source: outputs/sim/d53_experiment; decision log D-53; independent math review (21 Oct) §7. Provisional choice.

# Replaying the same evening fairly

**One synthetic night**

same bookings
same hidden patience and presence
same Lyft delays




**Only the rule changes, so differences come from the rule.**

**30** synthetic nights, each replayed six ways (not 180 independent nights)

**2 vans** 28 bookings in a one-hour night: 14 per van-hour vs April-derived 12.85, plus a 10-min burst

**5 vans** the proposed evening: designed, not yet run

ASSUMED

Provisional: 2-min ordering lag, riders accept Lyft, 8.9% no-show, placeholder patience, van travel × 1.2

Source: decision log D-56, D-57; outputs/sim/batch_L5_v1/summary.md.

# What our verification establishes

## Verification: the code does what we specified

**to the second**

3 designed evenings reproduce hand-computed answers

**292,181**

plan combinations on the designed evening, checked by an independent brute-force program over the same candidates; always agreed

**253**

automated tests pass (168 written by separate AI review agents that did not reuse the simulator's code)

**6 of 6**

sampled reruns byte-identical

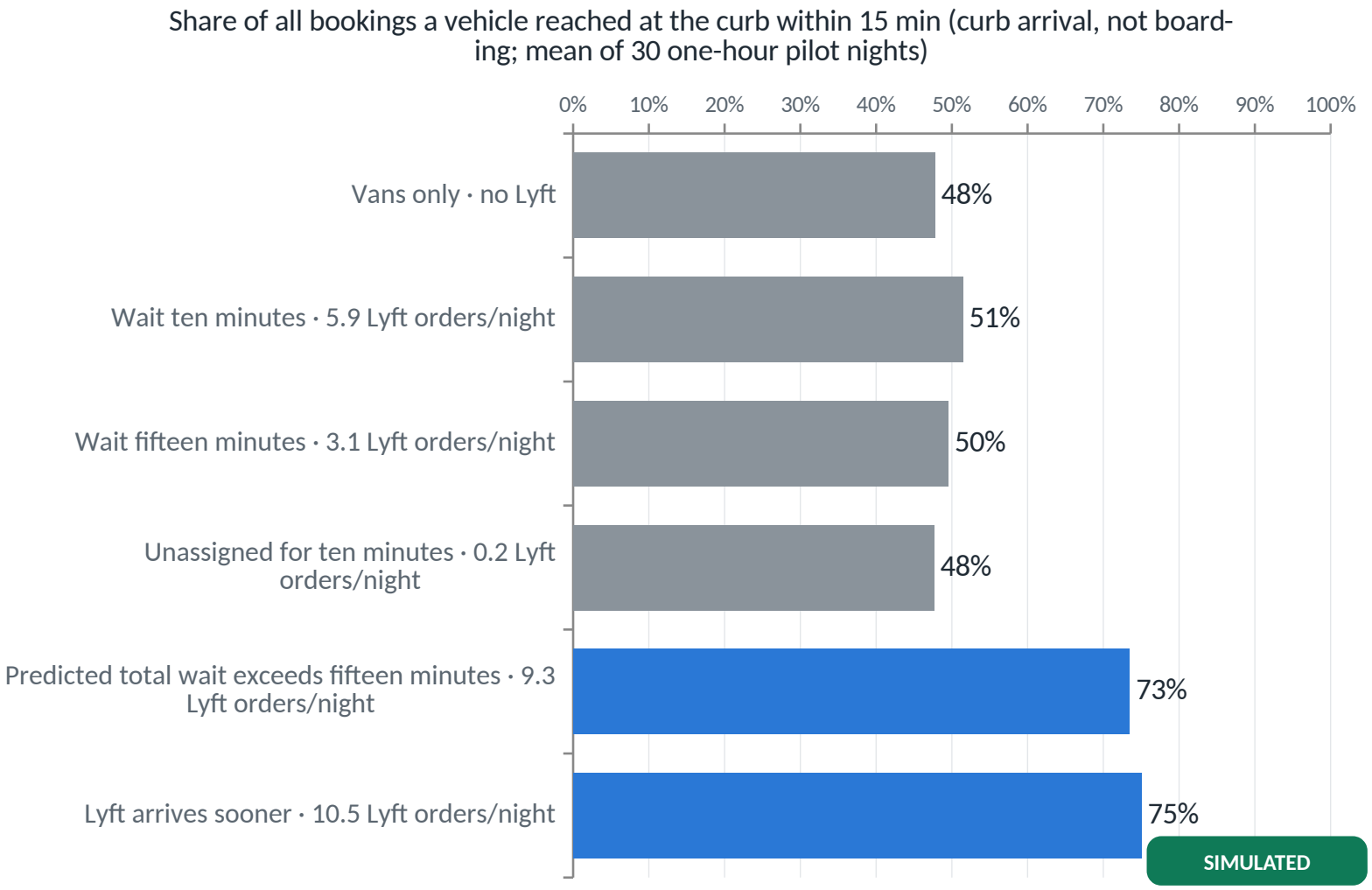
Numerical diagnostics: B8 →

## Not yet validation

- Do riders behave like our hidden traits?
- Does the real service match our evenings?
- Would these rules work on Homewood nights?
- These need data or trials we do not yet have

Source: Phase 4 log; tests/; outputs/sim/batch_L5_v1/reproducibility.json.

# Earlier prediction changed service and Lyft use




**Prediction rules**

reach about 73–75%, against 48–51%, using about 9–10.5 Lyft orders a night. They act before the long wait has happened.

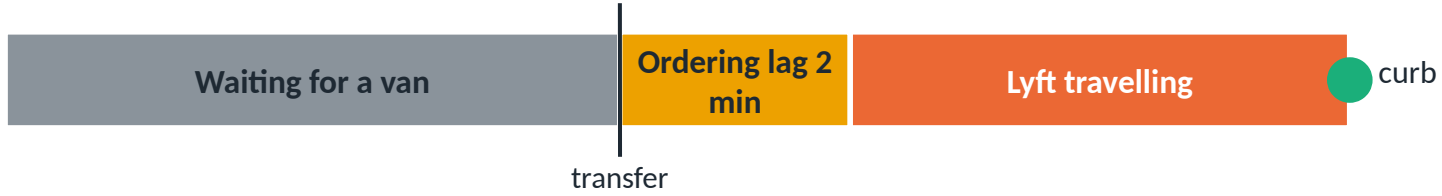
**Assigned is not picked up**

“Unassigned for ten minutes” was mostly inactive: identical to vans only on 24 of 30 nights; 8 transfers on 6 nights. Bookings were usually in some van's plan within minutes, even when pickup was far off; only during the burst did some stay unplanned for ten minutes.

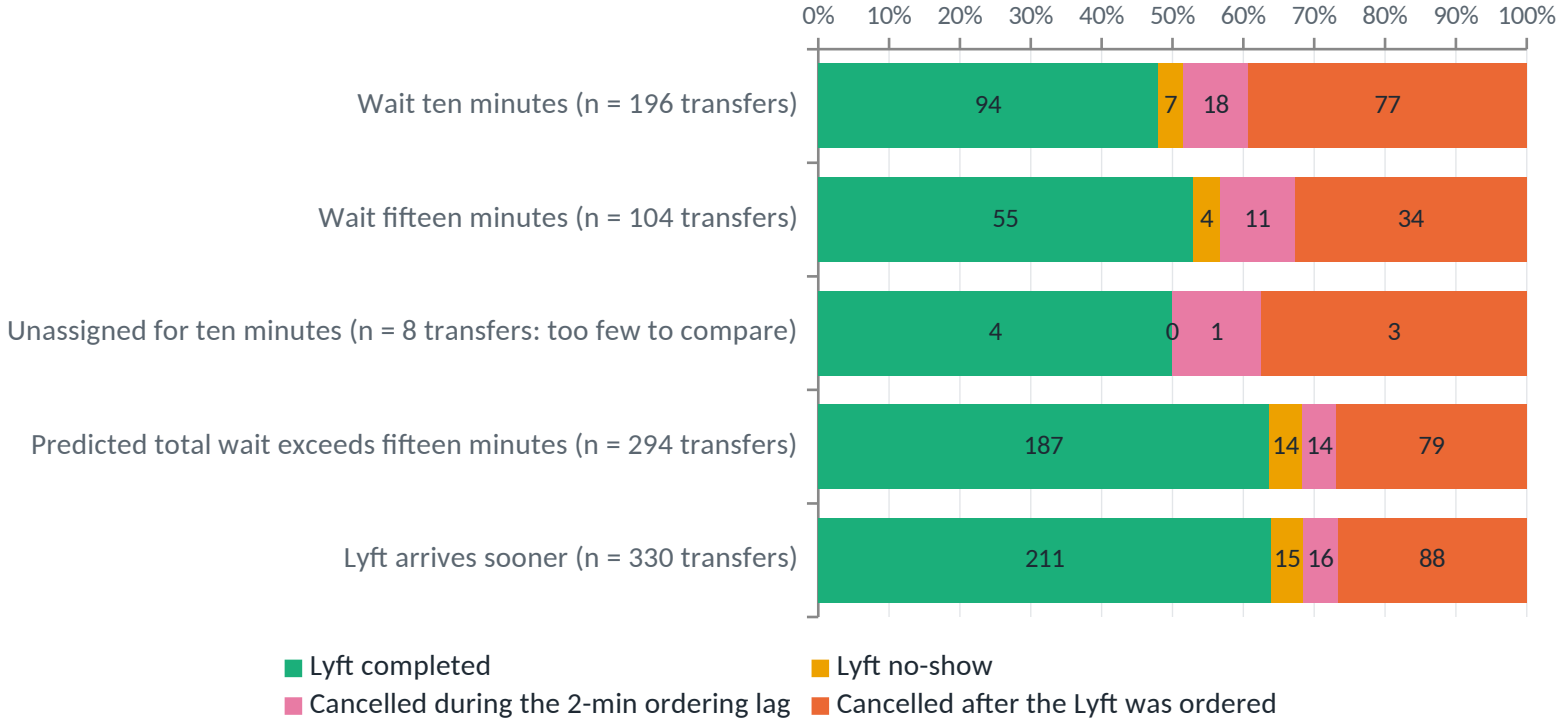
All metrics: B6 →

Source: outputs/sim/batch_L5_v1 (tradeoff.md, summary.md). Descriptive means; shares vary by SD 0.07–0.10 between nights; paired night-by-night counts in B7.

# A transfer does not end the rider's wait




*the rider's hidden patience clock keeps running the whole time*




**95 of 196**

bookings transferred under “Wait ten minutes” cancelled before the Lyft arrived (77 after the Lyft was ordered)

Prediction rules: about a third cancelled, consistent with earlier transfers leaving more patience for the Lyft (not isolated by an experiment).

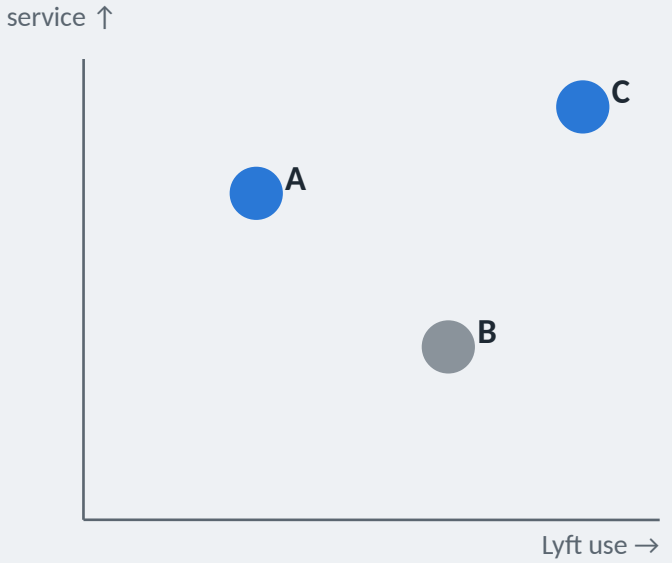
**ASSUMED**

Losses after transfer exist under any patience; their size and the gap between rules depend on the placeholder (impatience rising with time waited).

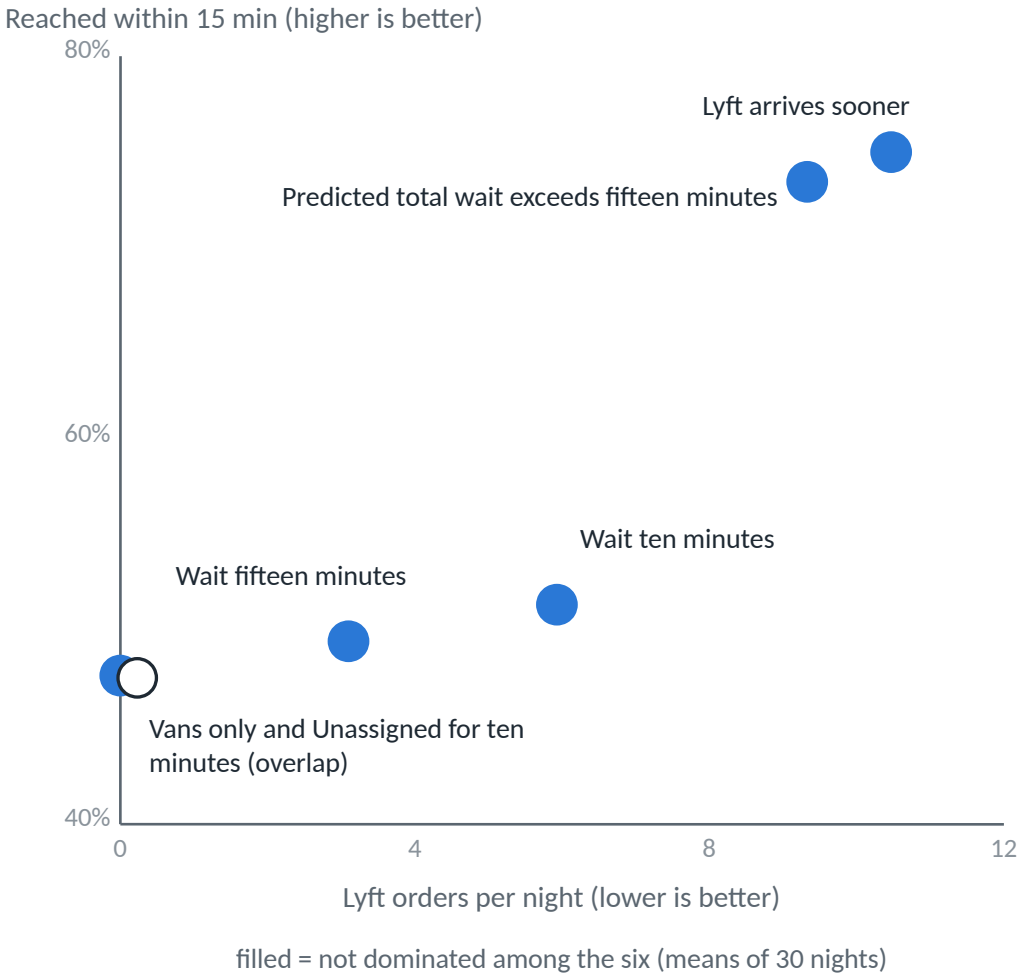
Source: outputs/sim/batch_L5_v1 per-run outcomes; totals over 30 nights; transfers = Lyft orders + cancellations during the ordering lag.

# Choosing between service and resource use

## Dominance, in one picture




A dominates B: at least as good on both, better on one. A and C: neither dominates.




filled = not dominated among the six (means of 30 nights)

- Two objectives: service up, Lyft orders down
- Five of six not dominated. “Unassigned for ten minutes” is dominated by vans only: about the same service, slightly more Lyft (0.23 vs 0 a night)
- Not dominated ≠ recommended: choosing needs a value on Lyft use
- Six tested rules, not a frontier of every rule
- Separate from the routing points of slide 12

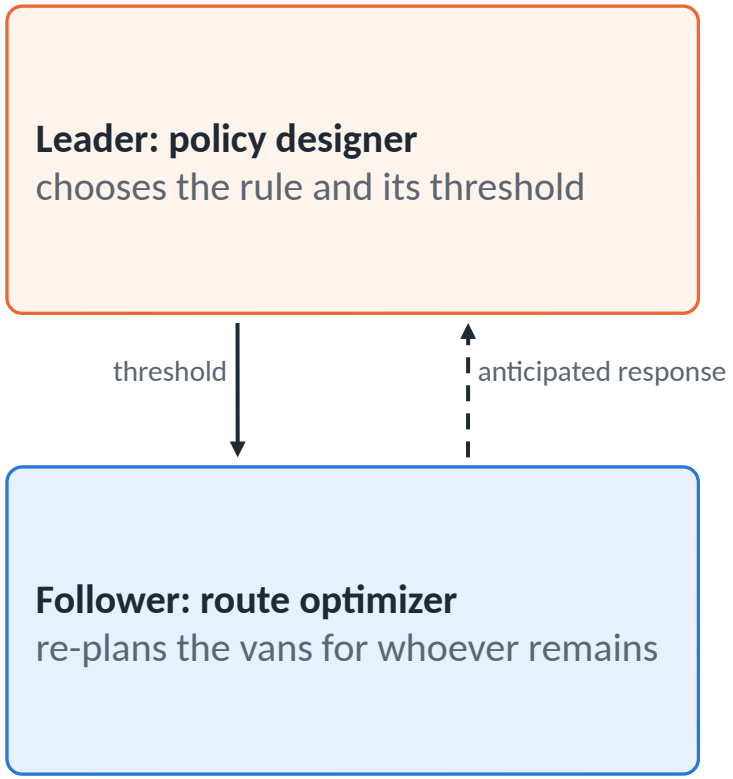
Night by night: B7 →

Source: src/sim/tradeoff.py; outputs/sim/batch_L5_v1/tradeoff.md. Descriptive, not a ranking.

# Could we optimize the transfer policy itself?

NOT IMPLEMENTED

A possible extension, not a result




choose $\theta$ to minimise service loss($\theta$) + $\lambda \cdot$ Lyft use($\theta$)
subject to: the vans' routes are the follower's optimal response, given $\theta$

## We have

- A simulator that evaluates any fixed rule
- Six fixed rules compared on identical nights
- Within each decision moment: routes chosen, then the rule, then at most one re-plan

## Still missing

- An optimization over the threshold $\theta$
- The follower's response written as constraints over the whole evening
- A treatment of uncertainty across nights
- An agreed value $\lambda$ for Lyft use
- (today the follower optimizes one decision moment at a time, over restricted candidates)

Source: methodology v1.4 §7b; Colson, Marcotte & Savard (2007); Dempe (2002); von Stackelberg (1934).

# What we learned and what must be tested next

**So far: deciding earlier, on a prediction, reached about 3 in 4 bookings within 15 min instead of about 1 in 2, at about 10 Lyft orders a night, in our two-van pilot. Whether that is worth it needs a value on Lyft use.**

## Learned

- Being assigned is not being picked up
- A transfer is not a completed trip
- Rules acting on predictions changed both service and Lyft use

## Limits

- Aggregate data only
- Placeholder patience
- Two-van pilot, not the five-van evening
- Verification, not validation
- 36 ride-time excesses across the 180 runs (16 distinct bookings on 10 nights), each coinciding with a no-show wait on that van (attribution, not proof)

## Next

- Agree how to judge the rules, then paired comparisons on the saved nights
- Sweep the thresholds
- Run the five-van evening and check solve times
- Test patience and Lyft-delay assumptions

Source: methodology v1.4 §7–§9; decision log D-57; outputs/sim/batch_L5_v1.

Backups →

# Backup slides

B1 Code key · B2 Route-selection MILP · B3 Event order · B4 Where inputs come from · B5 Three postponement prices · B6 All metrics by rule · B7 Night by night · B8 Numerical diagnostics · B9 Corrected proposal claims

B1 B2 B3 B4 B5 B6 B7 B8 B9

# B1  Code key (for files and questions only)

| Said in the talk | Code in files and logs |
|---|---|
| Vans only | R0 |
| Wait ten minutes | R1(10) |
| Wait fifteen minutes | R1(15) |
| Unassigned for ten minutes | R1u(10) |
| Predicted total wait exceeds fifteen minutes | R2(15) |
| Lyft arrives sooner | R3(2) |
| Small designed evenings with hand-computed answers | L1–L4 (L4: vans from 18:15, Lyft lag 2 min, delay fixed 8 min, acceptance 0.9) |
| The thirty-night pilot | L5: 30 synthetic one-hour nights, 2 vans, 28 bookings |
| The van optimizer | Dispatcher C: restricted route-selection MILP (HiGHS) |
| The postponement price we chose | Pending price A (earliest physical pickup) |
| Numbered decisions | D-xx entries in docs/technical/decision_log.md |

Source: src/sim/lyft.py; scenario_spec.md §6, §8; decision log.

← back to slide 13

# B2 The route-selection MILP at one decision moment

$$\min \ \Sigma_v \Sigma_k c_{vk} y_{vk} + \Sigma_r \pi_r u_r$$

$$\text{s.t.} \ \Sigma_k y_{vk} = 1 \quad \text{for every van } v \qquad \text{(one route per van)}$$

$$\Sigma_{\{v,k:\, r \in S_{vk}\}} y_{vk} + u_r = 1 \quad \text{for every waiting booking } r \quad \text{(served once, or postponed)}$$

$$y, u \in \{0, 1\}; \quad S_{vk} = \text{bookings picked up by candidate route } k \text{ of van } v$$

- $c_{vk}$: route points (waiting $\varphi$ with slopes 1/2.5/5/10/20 per passenger-minute after 0/5/10/15/20 min; riding 0.5 per passenger-minute; 0.1 per km); exact integers
- Candidates: every subset of at most 3 waiting bookings per van, each in its best stop order (exact search); the van's committed work is always a candidate
- $\pi_r = b_r + M$, $b_r$ from the chosen postponement price; $M = 1 + \Sigma_v (\max_k c_{vk} - \min_k c_{vk}) + \Sigma_r b_r$ guarantees every proven-optimal solution postpones the fewest bookings
- Ride-time guardrail max(25, 2 × direct + 5) min: no candidate may create or worsen an excess
- HiGHS: relative gap 0, absolute gap 0.5 units; a failed solve is logged, never replaced by a heuristic

Source: model_design.md v0.5 §3; decision log D-47, D-52 (amendments), D-53.

← back to slide 12

# B3  Event order and a booking's life

- Same-second order: drop-offs → curb arrivals (stop the patience clock) → service ends → vans on/off duty → new bookings → patience checks → Lyft bookings → planning → departures
- A van commits to a stop when it departs toward it; a booking with a van driving to it is not eligible for transfer
- After a transfer: a 2-min booking lag, then the Lyft's hidden delay; the rider may cancel at any point before the Lyft reaches the curb
- During an absent rider's 3-min curb wait, planning assumes possible boarding at expiry; the simulator never boards an absent rider
- Presence is one draw per rider, used whichever vehicle arrives
- Planning takes zero simulated time; computer time is logged only

Source: scenario_spec.md v0.8 §17; decision log D-46, D-48, D-54.

← back to slide 11

# B4 Where each input comes from

| Input | Status | Value | Basis |
|---|---|---|---|
| Bookings per hour | DERIVED | 64 / 50 / 54 for five vans; pilot: 28 in one hour, two vans | D-35, D-36, D-56 |
| Trip pairing | ASSUMED + fitted | Distance prior, decay 0.1 per min, fitted to both map totals | D-18 |
| Group size | ASSUMED (mean observed) | Shares 1–5: .73/.18/.055/.025/.01; April: 1.386 van passengers per completed van booking | D-38 |
| Patience | ASSUMED placeholder | Weibull, shape 1.5, median 18 min (scale ≈ 23.0 min) | D-39 |
| No-show | DERIVED, conditional | 8.9% per vehicle arrival if the status mapping holds (4.7% of all bookings) | D-40, D-48 |
| Lyft | ASSUMED | Ordering lag 2 min; delay lognormal, median 8 min, 90th percentile 15 min | D-43 |
| Travel times | DERIVED; × 1.2 ASSUMED | Local OSRM routing on the OpenStreetMap network; van times × 1.2 in the pilot (placeholder) | D-31, D-47 |

Source: decision log D-18–D-56; methodology v1.4 §4.

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# B5  Three postponement prices compared

| | First version | Chosen (earliest physical pickup) | Route-based |
|---|---|---|---|
| Price of postponing a booking | as if served at the next 2-min re-plan | as if served at the earliest time any on-duty van can reach it | as if served after a chosen van's route |
| Wait median / 90th pct / longest (min) | 17.1 / 31.0 / 44.4 | 18.6 / 31.2 / 35.6 | 21.1 / 34.3 / 38.6 |
| Mean wait (min) | 18.8 | 18.8 | 21.8 |
| Party of five from slide 15: wait | 22 min 18 s | 19 min 28 s | 20 min 39 s |
| Extra variables / solve time | 0 / 0.1 s total | 0 / 0.1 s total | up to 9,828 / 4.8 s total |
| Direction of error | optimistic | optimistic (ignores the van's other work) | either direction |

Designed evening, vans only. Chosen for its meaning and zero added size, provisionally; neither alternative guarantees a feasible schedule for all waiting riders.

Source: outputs/sim/d53_experiment/d53_comparison.md; decision log D-53; review d53_pending_price_review (22 Oct).

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# B6  All six rules over 30 synthetic nights

| Per night | Vans only | Wait ten minutes | Wait fifteen minutes | Unassigned for ten minutes | Predicted total wait exceeds fifteen minutes | Lyft arrives sooner |
|---|---|---|---|---|---|---|
| Reached within 15 min (share of 28) | 0.48 (0.09) | 0.51 (0.08) | 0.50 (0.09) | 0.48 (0.10) | 0.73 (0.07) | 0.75 (0.08) |
| Bookings picked up (of 28) | 16.1 (2.4) | 17.2 (2.6) | 16.5 (2.6) | 16.2 (2.5) | 20.2 (2.0) | 20.4 (2.3) |
| Wait median, min (per booking picked up) | 8.6 (1.8) | 8.3 (1.6) | 8.4 (1.5) | 8.9 (2.1) | 7.5 (1.0) | 7.3 (1.0) |
| Wait 90th pct, min (per booking picked up) | 20.2 (5.3) | 18.1 (2.4) | 19.6 (4.0) | 20.2 (4.6) | 13.3 (1.5) | 12.2 (1.7) |
| Wait 90th pct, min (per passenger picked up) | 20.1 (5.8) | 17.4 (2.7) | 19.2 (4.8) | 19.6 (4.7) | 13.2 (2.3) | 12.7 (2.7) |
| Van completed | 16.1 (2.4) | 14.1 (2.3) | 14.6 (2.4) | 16.1 (2.5) | 14.0 (2.7) | 13.3 (2.6) |
| Lyft completed | 0.0 (0.0) | 3.1 (1.4) | 1.8 (1.2) | 0.1 (0.3) | 6.2 (1.9) | 7.0 (1.6) |
| Cancelled waiting for van | 10.8 (2.3) | 6.5 (2.3) | 8.9 (2.5) | 10.5 (2.4) | 3.3 (1.7) | 2.6 (1.5) |
| Cancelled after transfer | 0.0 (0.0) | 3.2 (1.8) | 1.5 (1.0) | 0.1 (0.4) | 3.1 (1.5) | 3.5 (1.5) |
| Lyft orders | 0.0 (0.0) | 5.9 (1.6) | 3.1 (1.4) | 0.2 (0.5) | 9.3 (2.5) | 10.5 (1.9) |
| Van-km | 51.2 (4.4) | 47.6 (5.5) | 48.6 (5.0) | 51.9 (4.8) | 45.7 (5.7) | 43.0 (4.0) |
| Run time, s | 43 (46) | 40 (44) | 49 (57) | 50 (56) | 5 (7) | 4 (5) |

Cells: mean of nightly values (standard deviation across nights). Waits cover only bookings picked up. 30 nights replayed six ways, not 180 independent nights.

Source: outputs/sim/batch_L5_v1/summary.md.

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# B7  Trade-off night by night

| Rule | Reached (mean) | Lyft orders (mean) | Nights not dominated | Nights identical to vans only | Nights reach above / below vans only |
|---|---|---|---|---|---|
| Vans only | 0.477 | 0.00 | 30 of 30 | 30 | 0 / 0 |
| Wait ten minutes | 0.514 | 5.93 | 11 of 30 | 0 | 19 / 6 |
| Wait fifteen minutes | 0.495 | 3.10 | 14 of 30 | 1 | 15 / 5 |
| Unassigned for ten minutes | 0.476 | 0.23 | 26 of 30 | 24 | 2 / 2 |
| Predicted total wait exceeds fifteen minutes | 0.735 | 9.33 | 24 of 30 | 0 | 30 / 0 |
| Lyft arrives sooner | 0.750 | 10.47 | 12 of 30 | 0 | 30 / 0 |

A rule that makes no transfer on a night replays vans only exactly and ties it. Above / below counts are paired comparisons on the same synthetic night, with no intervals yet.

Source: outputs/sim/batch_L5_v1/tradeoff.md.

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# B8 Numerical diagnostics (thirty-night pilot)

- 180 runs = 30 synthetic nights × 6 rules; every run passed every check; one code version for all runs
- 13,663 decision moments (epochs); main solve proven optimal in all (none unproven, none failed)
- 41 epochs had an incomplete tie search (a solver error while looking for equally good plans); in all 41 a full enumeration shows a unique optimum, which is the plan chosen
- 36 ride-guardrail excesses across the 180 runs (16 distinct bookings on 10 nights, repeated across rules), each coinciding with a no-show wait on that van while the rider was aboard (attribution check, not proof); 0 unexplained
- Run time per run: mean 32 s, longest 253 s (Python, one core per run)
- Reproducibility: 6 of 6 sampled reruns identical byte for byte; the 2 batch runs of the first night identical to the earlier pilot outputs

Source: outputs/sim/batch_L5_v1 (summary.md, tie_incomplete_check.csv, reproducibility.json); decision log D-52 am. 5, D-57.

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# B9 Corrected proposal claims

| Proposal said | Corrected | Why |
|---|---|---|
| About 5,100 rides per hour at 6 pm | ≈ 170 completed rides in one evening's 6 pm hour | The bar is a 30-day total (tooltip 5,098; digitized 5,089) |
| About 28 vans | 27.6 vehicles on duty, “All services” average | Not a Homewood-only count |
| 1 in 4 trips overflow to Lyft | 24.5% of 44,374 riders rode Lyft | A share of riders, not of trips or requests |
| Trip-level records coming | Synthetic evenings built from aggregate dashboards | The records did not arrive |
| Compare dispatch algorithms | Compare transfer rules, van optimizer held fixed | On synthetic nights an algorithm comparison would mostly compare our assumptions |
| Lyft as paid overflow inside the optimizer | A separate, explicit transfer rule; no money modelled | A dispatcher sends the Lyft; the cost per Lyft trip is unknown |
| Targets as hard constraints | Reporting thresholds plus a ride-time guardrail | Hard targets would be stricter than the observed service; target wording changed over time |

The September proposal deck is unchanged; these corrections are recorded in methodology v1.4 §1a.

Source: methodology v1.4 §1a; decision log D-25, D-35, D-36, D-45, D-47, D-49, D-51.

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